

MOHAMED EBRAHIM

Data Analyst & ML Engineer

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Summary

AI and Data Science student at Kafr El-Sheikh University with hands-on experience in machine learning, NLP, and data analytics. Built end-to-end projects spanning deep learning models, RAG systems, and interactive BI dashboards. Completed internships at NTI and ITI (NVIDIA DLI), and seeking internship opportunities in Data Analysis, Machine Learning, or AI.

Experience

Information Technology Institute (ITI) — NVIDIA DLI

Nov 2025 – Dec 2025

Generative AI & LLM Applications Trainee

Remote, Egypt

- Completed training on Generative AI fundamentals: LLM architectures, Prompt Engineering, and RAG systems.
- Built retrieval-augmented pipelines to improve LLM response relevance and reduce hallucinations.
- Applied Python and NLP techniques in hands-on exercises and small-scale projects.

National Telecommunication Institute (NTI)

Jun 2025 – Jul 2025

Data Analysis Intern

Remote, Egypt

- Focused on applied data analysis techniques on telecom datasets using Python and SQL.
- Built and evaluated machine learning models for classification tasks.
- Generated insights and reports to support data-driven decision making.

Projects

Financial Market Pipeline | *ETL, HDFS, Spark, Airflow* | [GitHub](#)

Apr 2026

- Designed a containerized Big Data pipeline using Docker, Hadoop, Spark, and Airflow for data orchestration.
- Built a distributed ETL pipeline via Yahoo Finance API, HDFS, PySpark transform to Snowflake Star Schema DWH.
- Designed fault-tolerant Airflow DAGs with retry logic, scheduling to monitoring ingestion, transform, validation, loading
- Developed a Power BI dashboard with a Snowflake Star Schema model, enabling trend analysis, volume insights, and cross-asset performance tracking.

AI Faculty Regulations Assistant (RAG System) | *Python, Flask, TF-IDF, NLP, Ollama* | [GitHub](#)

Mar 2026

- Architected Arabic-first RAG system with 4-layer modular pipeline, cutting query resolution time from 3+ M to 5+ S
- Organized an Arabic-first Retrieval-Augmented Generation (RAG) system for answering university regulation queries.
- Designed a hybrid retrieval engine combining TF-IDF, keyword matching, and intent filtering to improve precision.
- Integrated local LLM (Qwen2.5) enabling fully offline inference with zero API cost.
- Deployed with Flask using a layered architecture, response caching, and explainable source cards.

Healthcare Analytics Dashboard | *Python, SQL, Power BI* | [GitHub](#)

Dec 2025

- Processed 10K+ healthcare records (patients, doctors, appointments, revenues) using Python and SQL.
- Built interactive Power BI dashboards tracking KPIs: attendance rates, revenue trends, and specialty breakdown.
- Organized into clean pipeline: raw data ingestion, Python scripts, SQL queries, and Power BI output.

Technical Skills

Languages: Python, SQL, C++

AI / ML: TensorFlow, Keras, Scikit-learn, PyTorch, XGBoost, Pandas, NumPy, OpenCV, CNNs, NLP, RAG, LLMs

Data & BI: Power BI, DAX, Excel, EDA, Data Cleaning, Data Visualization

Tools: Git, GitHub, VS Code, Flask, Jupyter, Google Colab

Volunteering

Student Activity

Spring 2023 – Spring 2024

Microsoft Student Club, IEEE, Mansoura Robotics Club

KSU

- Contributed to 3+ data-driven projects and AI/ML challenges, improving model accuracy by 15% using Python.
- Developed 5+ interactive Power BI dashboards, utilizing DAX to generate actionable insights from complex datasets.
- Optimized 10+ SQL queries, reducing data retrieval time by 25% and supporting efficient database design.
- Demonstrated strong analytical thinking and effective communication while collaborating within cross-functional teams.
- Integrated 2+ machine learning models into robotics systems, enhancing real-time decision-making and system performance.

Education

Kafr El-Sheikh University

2023 – 2027

Bachelor of Science in Artificial Intelligence